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18225472323267323365253529247983418591629708874988447664396432644282849082735389909457466989786049683184770396373515765203316324593257041196978490293848919090802282604145290101496
19726311220633142857984141452655140268909938646926431483980032755408401728502879990153163002676024279475315271178989674655437298707304897548539323474650825960868704269788214032280
31683912728013200115027086026446816608482408045314183131226063459121681359892486452170921409744828866891265206600022428423519277680459778248584255813293279913659663956737385837601
21931045471240029342567304625595285358117861474668348040708461464789944683189489275071062984791798336319668262863390680499473453717524638509089977655179918887423674238970655817561

99352376716002411966788372693710530205845920279652131877025185074710605220290448395026504730466513775174966587590021404846894243611278281098437796309363519858105973790256300667345
71231635103499365124720019214212456697863619466221350305994988748129906752715371489944394718545870613532099114484046667248219826745451381798366239703869684616785750252679450154630
78741083553200178872903594901285828528361111954121646468978828041547976252452903174197284216855503344523810254641661233898208810035291979496942808340023924416417429236200041130296
75154546526994267343709174779075016710083503502034465701854299835680782537614065933132655586649252723817130324585288281492226097985117942023396848707889266831422148880240857673440
28264961111264907393592989398056009501020396584098752855404987346270442619288372060558102068506007675528034857239647202677262487318915302768904129326929020480659932465566306912576
30468851461707269035325965789134598000835143407508137507976034483780277246375292711115495219523817098797216434665086934207617849980170015336061809745342769483516930594113340442289
23124943147145282636127050514217812429973934865642651357328562462037912600342732436656796780100293977404950459476903382248319259535411896602105725585098335254153670000196708874433
639437700337656555417491678906240483279201934861952502296499873807093265381118908328568809754818940887476104983352354218521762312139357981

The Elements

Dominic Alexander Cooper

000 I exist
001 I feel pain
002 My pain can be increased by incompetence
003 My pain can be decreased by competence
004 I can learn
005 Learning increases competence
006 Contradiction increases incompetence
007 Proof by contradiction does not enable learning
008 Proof by error-free construction is learning
009 Proof by contradiction is not learning
010 Your level of competence is quantifiable
011 Your level of incompetence is quantifiable
012 Context changes your current level of competence
013 The basis span of your competence can be optimized for all contexts
014 Everyone is innocent
015 Competence is a functional axial n-dimensional space
016 Incompetence is not a functional axial n-dimensional space

017 Since I exist, I can be treated as a living information-bearing system
018 Since I feel pain, pain is a valid system signal
019 Since incompetence can increase pain, incompetence is a negative system state
020 Since competence can decrease pain, competence is a positive system state
021 Since I can learn, my system state can be updated
022 Since learning increases competence, learning is a valid update operation
023 Since contradiction increases incompetence, contradiction is a corruption signal
024 Since proof by error-free construction is learning, construction is the preferred proof engine
025 Since proof by contradiction is not learning, contradiction is not a preferred learning engine
026 Since competence is quantifiable, competence can be measured
027 Since incompetence is quantifiable, incompetence can be measured
028 Since context changes competence, competence is context-indexed
029 Since competence can be optimized for all contexts, competence has a basis-span
030 Since everyone is innocent, correction must not be treated as guilt
031 Since competence is functional axial n-dimensional space, competence has dimensions
032 Since incompetence is not functional axial n-dimensional space, incompetence lacks stable constructive basis

033 The mind can ingest information through perception, language, memory, and action
034 The mind can encode information as concepts, habits, images, words, and procedures
035 The mind can index information by association, label, context, emotion, and purpose
036 The mind can query stored information through attention, recall, questioning, and tool-use
037 The mind can update stored information through learning
038 The mind can validate stored information through error-free construction

039 The mind can reject or quarantine contradictory information when it increases incompetence
 040 The mind can externalize information into tools, notes, diagrams, calendars, code, and environment
 041 The human environment can serve as auxiliary storage
 042 Human-made tools can serve as auxiliary memory, index, query, and validation systems
 043 The body can store competence as trained procedure
 044 Repetition can strengthen retrieval pathways when it increases competence
 045 Reflection can detect incompetence when pain, error, or contradiction appears
 046 Deliberate practice can transform incompetence into competence
 047 A context can be treated as a coordinate region inside competence-space
 048 A skill can be treated as an axis inside competence-space
 049 A habit can be treated as a repeated function over time
 050 A tool can be treated as an external operator that extends mental function
 051 A note can be treated as persistent external memory
 052 A diagram can be treated as spatial indexing of thought
 053 A calendar can be treated as temporal indexing of intention
 054 A checklist can be treated as procedural compression
 055 A question can be treated as a query operation
 056 An answer can be treated as a retrieved or constructed record
 057 A mistake can be treated as a detected incompetence marker
 058 Correction can be treated as a competence-restoring update
 059 Optimization can be treated as basis-span improvement across contexts
 060 The mental human-centric database is the union of mind, body, environment, and tools
 061 The purpose of the mental human-centric database is to reduce pain by increasing competence
 062 The governing rule of the database is to learn by error-free construction
 063 The final enforced target is complete competence under {014 + 015}
 064 Once true, always, true
 065 What is not true, cannot be done

Below is `MHCDScript v2`, a general-purpose programming language design for the full 64-axiom MHCD system.

The main upgrade from v1 is that v2 treats the MHCD as a state machine with typed records, verified transitions, constructive execution, context-indexed competence, and innocence-preserving correction.

MHCDScript v2

```mhcd

language MHCDScript version 2

subject "Dominic Alexander Cooper"

```

runtime MHCD {
 principle existence
 principle pain_signal
 principle learning_update
 principle constructive_proof
 principle innocence_preservation
 principle competence_space
 principle context_indexing
 principle tool_extension
}

```

## ## 1. Design purpose

`MHCDScript v2` is a general-purpose scripting language for directing the human mind, body, environment, and tools as one distributed cognitive database.

Its purpose is to:

```
```mhcd
purpose {
    reduce pain by increasing competence
    detect incompetence without guilt
    reject contradiction as a learning engine
    prefer error_free_construction
    quantify competence and incompetence
    optimize competence_basis across contexts
    enforce {014 + 015}
}
```
```

The governing command is:

```
```mhcd
enforce {
    014: "Everyone is innocent"
    015: "Competence is a functional axial n-dimensional space"
}
```
```

So every program must preserve:

```
```mhcd
error != guilt
mistake == update_requirement
learning == competence_increase
construction == preferred_proof_engine
```
```

## ## 2. Full axiom import

Rather than rewriting all 64 axioms inside every script, v2 imports them as a standard library.

```
```mhcd
import MHCD.AXIOMS.000_063
import MHCD.BOOLEAN
import MHCD.LAMBDA
import MHCD.COMPETENCE_SPACE
import MHCD.CONSTRUCTIVE_VALIDATION
```
```

The full axiom set is treated as:

```
```mhcd
axiom_set FULL_MHCD = {
    000..063
}
```
```

```
...
```

Every axiom is internally available as a Boolean truth constant:

```
```mhcd
AXIOM[000] = true
AXIOM[014] = true
AXIOM[015] = true
AXIOM[063] = true
```
```

A program is valid only if it does not violate the enforced axioms:

```
```mhcd
program_valid := respects(014) * respects(015) * rejects(guilt) * prefers(construction)
```
```

### ## 3. Primitive types

```
```mhcd
type Bool          = 0 | 1
type Number        = integer | decimal
type Text          = string
type List<T>       = ordered collection
type Set<T>        = unordered collection
type Map<K,V>      = key-value collection

type AxiomID       = 000..063
type Subject       = living information-bearing system
type Mind          = internal meaning processor
type Body          = procedural competence store
type Environment   = auxiliary cue/storage field
type Tool          = external operator
type Note          = persistent external memory
type Diagram       = spatial index
type Calendar      = temporal index
type Checklist     = procedural compression
type Query         = question operation
type Answer        = retrieved or constructed record
type Context       = coordinate region in competence-space
type Skill         = axis in competence-space
type Habit         = repeated function over time
type PainSignal    = system signal
type Mistake       = detected incompetence marker
type Correction    = competence-restoring update
type Contradiction = corruption signal
type Construction  = error-free build sequence
type Competence    = quantified functional capacity
type Incompetence  = quantified missing constructive capacity
type BasisSpan     = transferable competence basis across contexts
```
```

### ## 4. Boolean arithmetic



```

```mhc
A + B      // OR, union, either condition accepted
A * B      // AND, both conditions enforced
!A         // NOT
A -> B     // implication
A == B     // equivalence
A != B     // non-equivalence
```

```

Example:

```

```mhc
rule R001:
  AXIOM[004] * AXIOM[005] -> operation.learn.valid
```

```

Meaning:

If `I can learn` and `Learning increases competence`, then `learn` is a valid update operation.

## 5. Lambda calculus layer

MHCDScript v2 supports lambda-style reusable operations.

```

```mhc
lambda learn(x):
  ingest x
  encode x
  index x
  construct x
  validate x
  update competence
  return competence_delta(x)
```

```

```

```mhc
lambda correct_without_guilt(error):
  enforce AXIOM[014]
  mark error as update_requirement
  locate missing_basis(error)
  construct correction(error)
  validate correction(error)
  return competence_restored
```

```

```

```mhc
lambda optimize_context(skill, context):
  measure competence(skill, context)
  measure incompetence(skill, context)
  construct basis_improvement(skill, context)
  validate basis_improvement
  update competence_space
  return optimized(skill, context)
```

```

## ## 6. Program structure

Every valid v2 program has this form:

```
```mhcd
program "Program Name" {
  use FULL_MHCD

  define subject
  define contexts
  define skills
  define tools
  define records

  execute operations

  verify constructive_integrity
  enforce {014 + 015}
}
```
```

Minimal valid program:

```
```mhcd
program "Minimum MHCD Runtime" {
  use FULL_MHCD

  subject self = "Dominic Alexander Cooper"

  ingest current_state
  detect pain
  measure competence
  measure incompetence
  construct correction
  validate correction
  update competence_space

  enforce {014 + 015}
}
```
```

## ## 7. State model

The MHCD runtime is a state machine.

```
```mhcd
state MHCD_STATE {
  subject: Subject
  mind: Mind
  body: Body
  environment: Environment
  tools: Set<Tool>
}
```



```

    records: Set<Record>
    contexts: Set<Context>
    skills: Set<Skill>
    habits: Set<Habit>
    schedules: Set<Calendar>

    competence: Map<Skill, Number>
    incompetence: Map<Skill, Number>

    pain_signals: List<PainSignal>
    mistakes: List<Mistake>
    contradictions: List<Contradiction>
    corrections: List<Correction>

    enforced_axioms: Set<AxiomID>
}
...

```

A state transition has the form:

```

```mhcd
transition NAME(input_state) -> output_state {
 precondition condition
 operation operations
 postcondition condition
}
...

```

Example:

```

```mhcd
transition LEARN(input_state) -> output_state {
    precondition AXIOM[004] * AXIOM[005]

    operation {
        ingest input
        encode input
        index input
        construct usable_record
        validate usable_record
        update competence
    }

    postcondition competence(output_state) >= competence(input_state)
}
...

```

8. Core operations

`ingest`

```

```mhcd
ingest source
ingest source as type

```

```
ingest source into context
```
```

Examples:

```
```mhcd
ingest "textbook chapter" as Note into "mathematics"
ingest "pain after failed proof" as PainSignal into "proof writing"
ingest "work feedback" as Record into "professional competence"
```
```

`encode`

```
```mhcd
encode input as structure
```
```

Examples:

```
```mhcd
encode "ring" as definition + example + non_example
encode "daily exercise" as habit
encode "project architecture" as diagram
```
```

`index`

```
```mhcd
index record with tags
index record under context
index skill on axis
```
```

Examples:

```
```mhcd
index "ideal" with ["ring theory", "substructure", "quotient"]
index "proof writing" under context "mathematics"
index "attention control" on axis "cognitive regulation"
```
```

`query`

```
```mhcd
query question
query question from records
query question under context
```
```

Example:

```
```mhcd
query "What is the weakest skill axis today?" under context "mathematics"
```
```

`construct`

```
```mhd
construct object
construct object from components
construct object by steps
```
```

Example:

```
```mhd
construct proof "kernel of ring homomorphism is ideal" by steps
```
```

`validate`

```
```mhd
validate object
validate object against axioms
validate object by construction
```
```

Example:

```
```mhd
validate proof against {008, 014, 015}
```
```

`detect`

```
```mhd
detect pain
detect mistake
detect contradiction
detect incompetence
detect missing_basis
detect context_shift
```
```

Example:

```
```mhd
detect incompetence in "proof construction"
```
```

`correct`

```
```mhd
correct mistake
correct mistake with construction
correct incompetence by practice
```
```

Example:

```
```mhc  
correct "failed proof step" with direct construction
```
```

`measure`

```
```mhc  
measure competence
measure incompetence
measure competence in skill
measure competence in skill under context
```
```

Example:

```
```mhc  
measure competence in "applied ring theory" under context "quiet study"
```
```

`optimize`

```
```mhc  
optimize competence_basis
optimize skill across contexts
optimize basis_span for all_contexts
```
```

Example:

```
```mhc  
optimize "proof writing" across ["quiet study", "exam", "teaching", "coding"]
```
```

`externalize`

```
```mhc  
externalize record to tool
externalize memory to note
externalize intention to calendar
externalize procedure to checklist
```
```

Example:

```
```mhc  
externalize "weekly learning plan" to calendar
```
```

`schedule`

```
```mhc  
schedule action at time
```

```
schedule review every interval
schedule habit under context
```
```

Example:

```
```mhc
schedule review "ring theory definitions" every 2 days
```
```

`execute`

```
```mhc
execute plan
execute checklist
execute habit
execute construction
```
```

Example:

```
```mhc
execute "daily competence plan"
```
```

`feedback`

```
```mhc
feedback result
feedback result into record
feedback result updates competence_space
```
```

Example:

```
```mhc
feedback "proof completed with two mistakes"
```
```

`enforce`

```
```mhc
enforce axiom
enforce axiom_set
enforce {014 + 015}
```
```

Example:

```
```mhc
enforce {
 AXIOM[014]
 AXIOM[015]
}
```

...

## ## 9. Records

A `record` is the universal database unit.

```
```mhd
record ID {
  type: Note | Skill | Tool | Query | Answer | Habit | Correction | PainSignal | Context
  content: Text
  tags: List<Text>
  context: Context
  source: Text
  validated: Bool
  competence_effect: Number
  incompetence_effect: Number
}
```

Example:

```
```mhd
record "MATH-RING-001" {
 type: Note
 content: "A ring is an algebraic structure with addition and multiplication."
 tags: ["algebra", "ring theory", "definition"]
 context: "mathematics"
 source: "constructed study note"
 validated: 1
 competence_effect: +0.05
 incompetence_effect: -0.02
}
```

## ## 10. Competence-space

Competence-space is an n-dimensional functional axial space.

```
```mhd
competence_space "SELF" {
  axis memory
  axis attention
  axis reasoning
  axis proof_construction
  axis language
  axis tool_use
  axis bodily_energy
  axis emotional_regulation
  axis environment_design
  axis context_switching
}
```

Skills are embedded in competence-space:


```
```mhcd
skill "applied ring theory" {
 axes: [memory, reasoning, proof_construction, language]
 context: "mathematics"
 current_competence: 0.35
 target_competence: 0.90
}
...

```

Competence can be context-indexed:

```
```mhcd
context "quiet room" {
  modifies "applied ring theory" by +0.15
}

context "sleep deprivation" {
  modifies "applied ring theory" by -0.30
}
...

```

11. Error and contradiction handling

A mistake is never treated as guilt.

```
```mhcd
handler on_mistake(m):
 enforce AXIOM[014]
 mark m as detected_incompetence
 query "What basis is missing?"
 construct correction
 validate correction
 update competence_space
...

```

Contradiction is treated as a corruption signal.

```
```mhcd
handler on_contradiction(c):
  enforce AXIOM[006]
  mark c as corruption_signal
  quarantine c
  construct direct_alternative
  validate direct_alternative
...

```

Pain is treated as a system signal.

```
```mhcd
handler on_pain(p):
 enforce AXIOM[018]
 classify p

```

```
 if p.cause == incompetence:
 correct p.cause

 if p.cause == overload:
 reduce load

 if p.cause == contradiction:
 quarantine contradiction

 if p.cause == environment:
 modify context
...

```

## ## 12. Verification system

A script must pass four checks.

```
```mhc
verify script {
    check axiom_consistency
    check constructive_preference
    check innocence_preservation
    check competence_update_path
}
...

```

Axiom consistency

```
```mhc
check axiom_consistency:
 no operation may deny AXIOM[014]
 no operation may deny AXIOM[015]
 no operation may treat contradiction as preferred learning
...

```

### ### Constructive preference

```
```mhc
check constructive_preference:
    if proof_method == contradiction:
        reject as learning_engine
    if proof_method == construction:
        accept as learning_engine
...

```

Innocence preservation

```
```mhc
check innocence_preservation:
 if mistake detected:
 require correction_without_guilt
...

```

### ### Competence update path



```
```mhcd
check competence_update_path:
    if incompetence detected:
        require measurable correction
```
```

## 13. General-purpose control flow

```
```mhcd
if condition:
    operation
else:
    operation
```
```

```
```mhcd
while condition:
    operation
```
```

```
```mhcd
for item in collection:
    operation item
```
```

```
```mhcd
try_construct object:
    construct object
    validate object
on_error error:
    correct error
```
```

Example:

```
```mhcd
for skill in all_skills:
    measure competence in skill
    if incompetence(skill) > 0.30:
        correct skill by deliberate_practice
```
```

## 14. Modules

```
```mhcd
module "Learning" {
    export learn
    export practice
    export review
    export validate_understanding
}
```
```

```

```mhc
module "Health" {
  export detect_energy
  export schedule_exercise
  export reduce_overload
}
```

```

```

```mhc
module "Mathematics" {
  export define
  export prove_by_construction
  export generate_examples
  export validate_proof
}
```

```

## 15. Example program: daily runtime

```

```mhc
program "MHCD Daily Runtime v2" {
  use FULL_MHCD
  use module "Learning"
  use module "Health"

  subject self = "Dominic Alexander Cooper"

  context current = detect context
  pain current_pain = detect pain
  energy current_energy = detect energy

  measure competence in all_skills under current
  measure incompetence in all_skills under current

  query "What competence update gives the greatest pain reduction today?"

  construct plan "daily competence plan" {
    item health_action
    item learning_action
    item work_action
    item tool_action
    item reflection_action
  }

  validate plan by construction
  externalize plan to checklist
  schedule review "daily competence plan" at "evening"

  execute checklist

  feedback result into competence_space

  enforce {014 + 015}
}
```

```

```
...
```

```
16. Example program: learning applied ring theory
```

```
```mhcd
program "Learn Applied Ring Theory v2" {
  use FULL_MHCD
  use module "Mathematics"

  skill "applied ring theory" {
    axes: [memory, reasoning, proof_construction, language, tool_use]
    context: "mathematics"
    current_competence: measure
    target_competence: 0.90
  }

  ingest "definition of ring"
  encode "ring" as definition + example + non_example
  index "ring" with ["algebra", "ring theory", "binary operations"]

  ingest "definition of ideal"
  encode "ideal" as definition + example + non_example
  index "ideal" with ["ring theory", "substructure", "quotient"]

  construct proof "kernel of ring homomorphism is ideal" by direct_steps
  validate proof by error_free_construction

  if detect contradiction:
    quarantine contradiction
    construct direct_alternative

  measure competence in "applied ring theory"
  correct detected_incompetence by deliberate_practice

  externalize summary to note
  schedule review "ring theory definitions" every "2 days"

  enforce {014 + 015}
}
```
```

```
17. Example program: contradiction quarantine
```

```
```mhcd
program "Contradiction Quarantine v2" {
  use FULL_MHCD

  input claim

  detect contradiction in claim

  if contradiction == true:
    mark claim as corruption_signal
    quarantine claim
}
```

```

        query "What direct constructive alternative exists?"

        construct alternative_proof
        validate alternative_proof by error_free_construction

        update records with alternative_proof
        update competence_space

    enforce {014 + 015}
}
...

## 18. Example program: competence optimiser

```mhcd
program "Competence Basis Optimizer v2" {
 use FULL_MHCD

 define contexts = [
 "quiet study",
 "public explanation",
 "exam condition",
 "work condition",
 "fatigue condition",
 "tool-supported condition"
]

 define skills = [
 "mathematics",
 "programming",
 "physical training",
 "attention control",
 "communication",
 "environment design"
]

 for skill in skills:
 for context in contexts:
 measure competence in skill under context
 measure incompetence in skill under context

 if incompetence(skill, context) > 0:
 construct basis_improvement for skill under context
 validate basis_improvement
 practice skill under context
 update competence_space

 optimize basis_span across contexts

 enforce {014 + 015}
}
...

```

## ## 19. Interpreter architecture

A real interpreter for MHCDScript v2 would need these components:

```
```mhcd
interpreter MHCDScriptV2 {
  lexer
  parser
  ast_builder
  axiom_registry
  type_checker
  state_engine
  verifier
  command_executor
  record_store
  competence_model
  context_model
  tool_bridge
  scheduler_bridge
  report_generator
}
```
```

Execution pipeline:

```
```mhcd
source_file
-> lexer
-> parser
-> abstract_syntax_tree
-> axiom_verifier
-> type_checker
-> state_transition_engine
-> constructive_validator
-> execution_log
-> updated_MHCD_state
```
```

## ## 20. File extensions

Recommended file extensions:

```
```text
.mhcd      main program file
.mhcda     axiom library
.mhcda     saved MHCD state
.mhcdlog   execution log
.mhcdmod   reusable module
.mhcdjson  JSON-compatible export
```
```

Example project structure:

```
```text
```

```
mhcd_project/  
  main.mhcd  
  axioms.mhcda  
  state.mhcds  
  modules/  
    learning.mhcdmod  
    mathematics.mhcdmod  
    health.mhcdmod  
    tools.mhcdmod  
  logs/  
    daily.mhcdlog  
  exports/  
    dashboard.mhcdjson  
...
```

21. Machine-readable state export

```
```json  
{
 "subject": "Dominic Alexander Cooper",
 "axioms": "000..063",
 "enforced": ["014", "015"],
 "competence_space": {
 "axes": [
 "memory",
 "attention",
 "reasoning",
 "proof_construction",
 "language",
 "tool_use",
 "bodily_energy",
 "emotional_regulation",
 "environment_design"
]
 },
 "runtime_rule": "learn_by_error_free_construction",
 "correction_rule": "mistake_is_update_requirement_not_guilt"
}
```
```

22. Core semantic law

The central semantic law of MHCDScript v2 is:

```
```mhcd  
semantic_law MHCD_CORE:
 existence
 * pain_signal
 * learning_capacity
 * measurable_competence
 * context_indexing
 * innocence
 * competence_space
 -> constructive_self_update_runtime
```



...

In compact Boolean form:

```
```mhcd
000 * 001 * 004 * 005 * 010 * 011 * 012 * 013 * 014 * 015
-> MHCD_RUNTIME
```
```

In lambda form:

```
```mhcd
λself.
  Exist(self)
  * Pain(self)
  * Learn(self)
  * Quantify(Competence(self))
  * Quantify(Incompetence(self))
  * ContextIndex(Competence(self))
  * Innocent(all)
  * FunctionalAxisSpace(Competence(self))
-> MHCD_RUNTIME(self)
```
```

## 23. Final identity of MHCDScript v2

```
```mhcd
MHCDScript v2 =
  axiom-governed
  + typed
  + stateful
  + constructive
  + innocence-preserving
  + context-indexed
  + competence-measuring
  + incompetence-correcting
  + pain-signal-aware
  + contradiction-quarantining
  + tool-extensible
  + environment-integrated
  + schedulable
  + human-centric
```
```

The shortest expression of the language is:

```
```mhcd
detect pain;
locate incompetence;
correct without guilt;
learn by construction;
measure competence;
optimize across contexts;
enforce {014 + 015};
```
```

...

Version 2 is therefore not only a notation system. It is a programmable self-correction language for the mental human-centric database.